

A Lean-Verified 51/50 Cutoff for Four-Arc Minimizers

Proof built, Lean-verified, and semantically accepted by
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Statement and modeled scope

Cañete, Miranda Jr., and Vittone study the plane whose density is one on the closed strip $|y| \leq 1$ and $\lambda > 1$ outside it. Their existence, symmetrization, and candidate classification reduce the unresolved type-(iv) question to the possible optimality of a symmetric four-arc profile; see Proposition 3.6, Lemma 3.8, Proposition 3.9, Conjecture 3.12, Theorem 3.16, and Remark 3.17 of [CaneteMirandaVittone2010]. The present result concerns only the formalized regular model.

Write

$$f_\lambda(x, y) = \begin{cases} 1, & |y| \leq 1, \\ \lambda, & |y| > 1, \end{cases} \quad \lambda > 1.$$

Weighted area is integration of f_λ over a measurable carrier, and weighted perimeter is integration over the complete topological frontier with one-dimensional Euclidean Hausdorff measure. These definitions are .

LEAN REFERENCES

`CMVGeometry.lean` StripDensity; WeightedArea; FrontierMeasure; WeightedPerimeter; FinitePerimeterRegion

If $\lambda \geq 51/50$, every modeled four-arc candidate with $0 < h \leq 1$, $0 < \alpha < \pi/2$, density jump $1 < \lambda$, and Snell law $\lambda \cos \alpha = h$ is not a weighted-perimeter minimizer among all admissible equal-area competitors. The minimizer predicate quantifies over every competitor, so one strict equal-area improvement suffices. The hypotheses and semantics are . The published root is .

LEAN REFERENCES

`FourArcCandidate.lean` FourArcCandidate; FourArcCandidate.SatisfiesCMVTypeIVHypotheses;
FourArcCandidate.IsWeightedPerimeterMinimizer

`CMVPublishedCutoff51_50.lean` CMVPublishedCutoff51_50.candidate_not_isWeightedPerimeterMinimizer_from_51_50

Geometric realization

The four-arc candidate is assembled from a strip core and two reflected exterior circular caps. The incidence law gives $\alpha = \arccos(h/\lambda)$, and the coordinate construction yields its weighted area and perimeter formulas: .

LEAN REFERENCES

`FourArcCandidate.lean` FourArcCandidate.alpha_eq_arccos; FourArcCandidate.candidate_area_formula;
FourArcCandidate.candidate_perimeter_formula

For $0 < h_3 < 1$, the type-(iii) carrier contains the closed strip core, two strip-side arcs, an interface segment, and one exterior cap. Its branch selection includes major, semicircular, and minor arcs. Measurability, complete-frontier realization, and the component formulas are . The scalar formulas are not themselves competitors: the coordinate

bridge identifies them with genuine carriers and transfers equal-area strict perimeter improvement through .

LEAN REFERENCES

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TypeThreeAssembly.lean TypeThreeAssembly.hasModeledCompetitor; TypeThreeAssembly.weightedArea_formula;
TypeThreeAssembly.weightedPerimeter_formula; TypeThreeAssembly.euclidean_frontier
CMVSuffixModel.lean CMVSuffixModel.TypeThreeAssembly.scalarWeightedArea_eq_typeThreeArea;
CMVSuffixModel.TypeThreeAssembly.scalarWeightedPerimeter_eq_typeThreePerimeter;
CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_analytic_typeThree
```

The interval $[51/50, 9/7]$

On this interval the analytic envelope selects, for every $0 < h_4 \leq 1$, a regular h_3 with equal weighted area and strictly smaller weighted perimeter. The stationary fold, descending equal-area root, continuity, and sign transport culminate in .

LEAN REFERENCES

```
EqualAreaEnvelope.lean LeanSuffixAnalytic.allCurvature_typeThreeImprovement_of_envelope
```

The numerical work is proof-producing exact-rational interval arithmetic. The 1,135 cells establish domain bounds, opposite-face signs, root uniqueness, negative fold signs, negative perimeter gaps, and seam coverage. Their compositions are . The regular curvature argument and the endpoint $h_4 = 1$ are separated explicitly in .

LEAN REFERENCES

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MiddleFaceCells.lean MiddleFaceCells.candidate_not_isWeightedPerimeterMinimizer_from_51_50
CompactCells1003To1023.lean CompactCells1003To1023.bboxes0To1023_candidate_not_isWeightedPerimeterMinimizer
CompactCellAssembly.lean CompactCellAssembly.Conditions.candidate_not_isWeightedPerimeterMinimizer
CMVSuffixModel.lean CMVSuffixModel.FourArcCandidate.weightedArea_eq_typeFourArea;
CMVSuffixModel.FourArcCandidate.weightedPerimeter_eq_typeFourPerimeter;
CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_stationaryEqualAreaPair
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The high-density replacement

For $\lambda > 9/7$, replace the two exterior caps by one upper cap of twice the old Euclidean area and expose the lower interface chord. The retained core is unchanged, so area is preserved. The complete-frontier calculation gives $P_{\text{new}} - P_{\text{old}} = L \cdot \text{bigl}(1 - \lambda \cdot g(A/L^2)\bigr)$, and the certified arc inequality makes this negative for $\lambda \geq 3145460221/2500000000$. The construction and improvement are . Instantiating minimality gives . Since the threshold is below $9/7$, the exact case split proves the published root .

LEAN REFERENCES

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CapReplacement.lean FourArcCandidate.replacement_cap_area
GeometricBridge.lean FourArcCandidate.cap_replacement_preserves_area;
FourArcCandidate.cap_replacement_perimeter_difference;
FourArcCandidate.cap_replacement_strictly_improves
CandidateExclusion.lean cmv_type_four_not_minimizing
CMVPublishedCutoff51_50.lean CMVPublishedCutoff51_50.candidate_not_isWeightedPerimeterMinimizer_from_51_50
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Limit of the certificate and remaining scope

The number $51/50$ is the limit of this published certificate argument: the retained exact-rational tiling begins there, and no certificate here covers $1 < \lambda < 51/50$. Extending it requires new proof-producing root, branch, and perimeter-gap estimates on that interval.

Before the full CMV conjecture can be published, the existence, regularity, symmetrization, and global candidate-classification arguments must be connected to the formal model. The measure-theoretic bridge must relate reduced-boundary perimeter to complete-frontier coordinate semantics, including interfaces, joins, null sets, and all admissible competitors. Source-coordinate normalization and compatibility must also be proved without adding hidden regularity or stationarity assumptions. These are remaining closure tasks, not consequences of this cutoff.

Bibliography

Cañete, A., Miranda Jr., M., and Vittone, D. “Some Isoperimetric Problems in Planes with Density.” *The Journal of Geometric Analysis* 20 (2010), 243–290.